COMP 2711 – Mock Midterm 09

Solutions and marking notes

Practice material – not an official examination

Problem 1 [10 points]

(a) **[3 points]**

$$((p \to q) \to p) \equiv \neg(\neg p \vee q) \vee p \equiv (p \wedge \neg q) \vee p \equiv p.$$

The laws are implication, De Morgan/double negation, and absorption. Therefore the biconditional reduces to $p \leftrightarrow p$, always T.
Marking: 1 implication expansion; 1 De Morgan/absorption; 1 distinction between equivalence and tautological biconditional.

(b) **[2 points]** Let p, q mean A,B are knights. Then $p \leftrightarrow \neg q$ and $q \leftrightarrow (p \vee q)$. If $q = F$, the first gives $p = T$, making B's assertion true, impossible for a knave. Thus $q = T, p = F$. A's assertion is false and B's is true, so A is a knave and B a knight.
Marking: 1 consistency reasoning; 1 unique assignment and verification.

(c) **[5 points]**

1. $p \to r$ from the first two premises by hypothetical syllogism.
2. p from $p \vee t$ and $\neg t$ by disjunctive syllogism.
3. r from lines 1 and 2 by modus ponens.
4. $r \to s$ from $\neg r \vee s$ by implication equivalence.
5. s from lines 3 and 4 by modus ponens.

Marking: 1 per valid derived line and rule citation.

Problem 2 [13 points]

(a) **[4 points]** (i) $\forall d \exists t(D(t,d) \wedge \forall x A(x,t))$. (ii) $\exists x \forall d \exists t(A(x,t) \wedge D(t,d))$. The first gives a shared test for a given device; the second gives a shared technician across devices.
Marking: 2 each: 1 quantifier dependency; 1 correctly scoped predicates.

(b) **[4 points]**

$$\forall x\big(\neg P(x) \vee \exists y(\neg Q(y) \wedge \forall z \neg R(x,y,z))\big).$$

Negation changes the outer conjunction to disjunction and the inner disjunction to conjunction while reversing quantifier types.
Marking: 1 x; 1 outer De Morgan; 1 y and inner De Morgan; 1 z and correct atomic form.

(c) [**5 points**] For the refutation use $D = \{0, 1\}$ and $R(x, y)$ iff $x = y$. Each x has itself as witness, but no y equals both 0 and 1. For the valid conclusion take an element a (possible because the domain is nonempty), instantiate to $\exists y R(a, y)$, and choose fresh witness b with $R(a, b)$. Existentially generalize first to $\exists x R(x, b)$ and then to $\exists y \exists x R(x, y)$.
Marking: 2 countermodel and values; 1 nonempty-domain selection/UI; 1 fresh existential witness; 1 two existential generalizations.

Problem 3 [12 points]

(a) [**2 points**] For $x \in U$, $x \notin A \cap B$ iff not both memberships hold, iff $x \notin A$ or $x \notin B$, iff x belongs to the right side. Elements outside U belong to neither side. Thus membership agrees everywhere.
Marking: 1 correct De Morgan membership equivalence; 1 universe restriction and equality.

(b) [**3 points**] Set $h = f^{-1} \circ g^{-1} : C \to A$. For $a \in A$, $h(g(f(a))) = f^{-1}(g^{-1}(g(f(a)))) = a$. For $c \in C$, $g(f(h(c))) = g(f(f^{-1}(g^{-1}(c)))) = c$. Thus h is the two-sided inverse; the inverse order is reversed.
Marking: 1 correctly typed candidate; 1 each order identity.

(c) [**7 points**]

(i) **T**. The bijection is $n \mapsto 2(n + 1)$.

(ii) **F**. Its power set has $2^5 = 32$ elements.

(iii) **T**. Countable includes finite sets and countably infinite sets.

(iv) **T**. List by the last possible one-position; each layer has finitely many sequences.

(v) **F**. Uncountability alone says nothing about the nonexistence of intermediate cardinalities; the lecture states CH is independent of ZFC.

(vi) **T**. The diagonal proof holds for every set, including the empty set.

(vii) **F**. Take $A = \{0\}$, $B = \{0, 1\}$; an injection exists but no bijection does.

Marking: Award 1 point per correct T/F choice (7 total); reasons are included here for checking.

Problem 4 [16 points]

(a) [**2 points**] The first two choices have $3 \cdot 5$ possibilities; six symbols then remain, followed by five: $3 \cdot 5 \cdot 6 \cdot 5 = 450$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(b) **[2 points]** Choose the common value in four ways and the different third value in three: $4 \cdot 3 = 12$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(c) **[2 points]** The unique three-one block may start at positions 1 through 8, giving 8 strings.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(d) **[2 points]** The unrestricted count is $11!/(4!4!2!) = 34650$. Compressing an adjacent PP pair into one block gives $10!/(4!4!) = 6300$ forbidden strings. Subtracting yields $34650 - 6300 = 28350$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(e) **[2 points]** Subtract assignments using only shelves 2 and 3: $3^7 - 2^7 = 2059$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(f) **[2 points]** Shift the first two variables down by their lower bounds. The residual total is five, giving $\binom{7}{2} = 21$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(g) **[2 points]** The coefficient is $\binom{8}{5} - \binom{7}{5} = \binom{7}{4} = 35$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(h) **[2 points]** Fix the winner. The final game is its third win; choose its other two wins among the previous $\ell - 1$ games. Thus the counts are $2\binom{2}{2} = 2$, $2\binom{3}{2} = 6$, $2\binom{4}{2} = 12$, totaling 20. No earlier third win is possible in such a prefix.
Marking: 1 mark for correct stopping-aware counts by length; 1 for total 20.

Problem 5 [10 points]

(a) **[6 points]** Count ordered pairs (R, S) of disjoint subsets of a fixed n-element set, with sizes r, s. Choose R first and then S from the remaining elements, giving the left side. Alternatively choose the union $U = R \cup S$ of size $r + s$, then choose the r members of R inside it; $S = U \setminus R$ is forced. This gives the right side. The assignments are inverse descriptions of each ordered pair, even if a group is empty.
Marking: 1 mark for disjoint labelled groups; 2 for the first count; 2 for the union-and-split count; 1 for the inverse correspondence/boundaries.

(b) **[4 points]** Label the ground-set elements $1, \ldots, n$. At position i, write R if $i \in R$, S if $i \in S$, and U otherwise. The disjointness makes the symbol well defined. Conversely the symbol positions recover both sets, so this is a bijection. Equal symbols are indistinguishable, and the common count is

$$\frac{n!}{r!s!(n-r-s)!}.$$

This also counts allocation of n distinct labelled objects into the three labelled categories of these sizes.
Marking: 2 marks for the explicit encoding and inverse; 1 for indistinguishable symbol interpretation; 1 for the factorial formula.

Problem 6 [9 points]

[9 points] Use four bad events: A in seat 1, A in seat 2, B in seat 3, and C in seat 4. Each has 5! seatings. The two A-events cannot overlap. Of the six pairs of events, the other five have compatible distinct people and seats, each giving 4!. Of the four triples, only the two formed by one A-event and both B,C events are compatible, each giving 3!. The quadruple is empty. IE gives

$$6! - 4 \cdot 5! + 5 \cdot 4! - 2 \cdot 3! = 348.$$

Marking: 2 marks for four bad events; 1 for singleton counts; 3 for compatible and incompatible pair/triple counts; 1 for empty quadruple; 2 for IE and answer.

Problem 7 [7 points]

[7 points] Pair consecutive coordinates in each of the three directions, partitioning the 64 unit cubes into $2 \cdot 2 \cdot 2 = 8$ disjoint $2 \times 2 \times 2$ blocks. Nine selected unit cubes force two in one block. All eight unit cubes of that block have its central point as a vertex; in particular the two selected cubes share that vertex.
Marking: 3 marks for the eight-block partition; 2 for objects/boxes and PHP; 2 for verifying the shared central vertex.

Problem 8 [6 points]

(a) **[1 points]** All ordered token pairs, including repeats, form the equally likely space of size $4^2 = 16$.
Marking: 1 mark for ordered pairs with repeats and 16.

(b) **[1 points]** Four diagonal pairs qualify among sixteen, giving $1/4$.
Marking: 1 mark for $1/4$.

(c) **[1 points]** $P(E \cap F) = (1/2)^2 = 1/4$.
Marking: 1 mark for $1/4$.

(d) **[1 points]** The disjoint orders red–blue and blue–red each have mass $1/4$, giving $1/2$.
Marking: 1 mark for the disjoint two-case sum.

(e) **[2 points]** The joint probability becomes $(2/4)(1/3) = 1/6$. Both marginals remain $1/2$, so $1/6 \neq 1/4$ establishes dependence.
Marking: 1 mark for the without-replacement probability; 1 for the independence comparison.

Problem 9 [7 points]

(a) **[3 points]** The likelihoods are $1, 1/8, 1/64$. Their prior-weighted sum is $P(R) = (1 + 1/8 + 1/64)/3 = 73/192$. Normalizing the weights $64, 8, 1$ gives posteriors $64/73, 8/73, 1/73$.
Marking: 1 mark for the conditional likelihoods; 1 for the total; 1 for all normalized posteriors.

(b) **[4 points]** Given the device, the fourth trial remains independent of its first three. Total probability with the updated device partition yields

$$P(S_4 \mid R) = \frac{64}{73} \cdot 1 + \frac{8}{73} \cdot \frac{1}{2} + \frac{1}{73} \cdot \frac{1}{4} = \frac{273}{292}.$$

Before observations the success probability is $(1 + 1/2 + 1/4)/3 = 7/12$, smaller than $273/292$. Evidence shifts mass toward more reliable devices.
Marking: 1 mark for conditional independence on device; 1 for the posterior-weighted sum; 1 for $273/292$; 1 for the prior comparison.

Problem 10 [10 points]

(a) **[2 points]** $E(X) = 6/4 = 3/2$ and $E(X^2) = 14/4 = 7/2$, so $V(X) = 7/2 - 9/4 = 5/4$.
Marking: 1 mark for the mean; 1 for the variance.

(b) **[2 points]** $E(Y) = E(X^2) = 7/2$, whereas $E(X)^2 = 9/4$. Their difference is $5/4$, the variance.
Marking: 1 mark for each quantity and their distinction.

(c) **[2 points]** The values are $0, 0, 2, 6$, so the mean is $8/4 = 2$. Independently, $E(X^2 - X) = 7/2 - 3/2 = 2$. The factors are dependent, so their expectations should not be multiplied.
Marking: 1 mark for the outcome average; 1 for the linearity computation.

(d) **[4 points]** Since $E(aX + b) = aE(X) + b$,

$$V(aX + b) = E\big([aX + b - aE(X) - b]^2\big) = a^2 E\big([X - E(X)]^2\big) = a^2 V(X).$$

For $X - Y$, the four values are $0, 0, -2, -6$, so its mean is -2 and second moment is 10. Its variance is $10 - (-2)^2 = 6$.
Marking: 2 marks for centered deviations and squared scaling; 1 for the transformed moments; 1 for variance 6.